

Influencer Fraud Detection - Technical Validation Report (v2)

Generated 2026-08-12 | Auto-generated from run_results.json - figures below are pulled directly from the pipeline run, not typed by hand.

1. Methodology Corrections vs. v1

- No dummy-label fallback: the pipeline refuses to run if real fraud labels are missing, instead of silently validating against random noise.
- No threshold leakage: contamination/threshold are grid-searched on a TUNE split only; all metrics below are computed on a held-out TEST split the search never saw.
- Threshold selection is 5-fold cross-validated within the tune split and requires a minimum number of flagged points, preventing the search from picking an extreme threshold that 'wins' by isolating one or two lucky points.
- A supervised baseline (Random Forest) was trained on the same tune split and evaluated on the same test split, so the unsupervised ensemble is judged against an honest comparison point instead of being assumed adequate.

2. Validation Setup

Total scored profiles: 5,000 | Labeled sample: 800 (tune: 400, held-out test: 400) | Grid-search-selected contamination: 0.05, threshold: 40.0

3. Results on Held-Out Test Split (never used for tuning)

Metric	Unsupervised Ensemble (IsolationForest + LOF)	Supervised Baseline (Random Forest)
Precision	0.500	1.000
Recall	0.205	0.977
F1 Score	0.290	0.989
ROC-AUC	0.915	1.000

Unsupervised ensemble confusion matrix on test split (rows=actual, cols=predicted): TN=347, FP=9, FN=35, TP=9.

4. Honest Finding & Production Decision

The unsupervised ensemble did **not** reliably clear the 85% precision bar on cross-validated tuning data, and its held-out test precision/recall (0.50 / 0.20) confirms it is not precise enough to drive rejection decisions on its own, despite a respectable ROC-AUC of 0.92 (it ranks risk reasonably well, it just doesn't threshold cleanly at the volumes required). The supervised Random Forest, trained on the same labeled data, reached 1.00 precision / 0.98 recall on the same unseen test split.

Production model selected: supervised rf plus anomaly flag. The supervised model drives the fraud_risk_score and vetting_action for all 5,000 profiles (refit on the full 800 labeled rows after test metrics were locked in). The unsupervised ensemble is retained as a secondary anomaly signal: any profile the supervised model scores as low-risk but the ensemble flags as a strong anomaly is marked with a novel_pattern_flag, since a supervised model can only recognize fraud patterns that resembled

something in its 800 labeled examples. This run raised 3 such flags.

5. Why This Differs From the v1 Report

The original submission reported precision=1.000 at recall=0.020 by tuning the threshold and reporting the final metric on the same 200 rows -- an extreme threshold that isolated a handful of points can look perfect and still be useless in practice. It also silently fabricated random fraud labels if none were supplied, and a companion report recommended a different threshold (65) than the one the technical run actually used (40), because the two documents were written independently rather than generated from one run. Both problems are structurally impossible in this version: the reported numbers here are computed once, in `run_results.json`, and both this report and the business report read from that same file.